

# *# Atmospheric Dynamics*



## *Chapter 4: Balanced Flow*



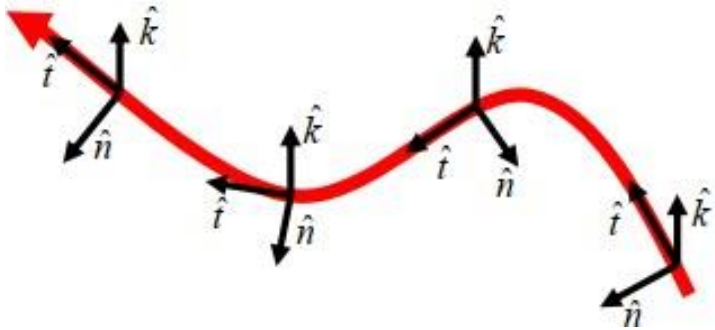
### *Objective*

- *To Identify Hydrostatic and geostrophic balance*
- *To Recognise Inertial flow and cyclostrophic flow*
- *To Know The gradient wind approximation*

- The pressure and velocity distributions in atmospheric systems are related by relatively simple, approximate ***force balances***.
- We can gain a qualitative understanding by considering ***steady-state*** conditions, in which the fluid flow does not vary with time, and by assuming there are ***no vertical motions***.
- To explore these balanced flow conditions, it is useful to define a new coordinate system, known as ***natural coordinates***.

# Natural Coordinates

- *Natural coordinates* are defined by a set of mutually orthogonal unit vectors whose orientation depends on the direction of the flow.



Unit vector  $\hat{t}$  points along the direction of the flow.

Unit vector  $\hat{n}$  is perpendicular to the flow, with positive to the left.

Unit vector  $\hat{k}$  points upward.

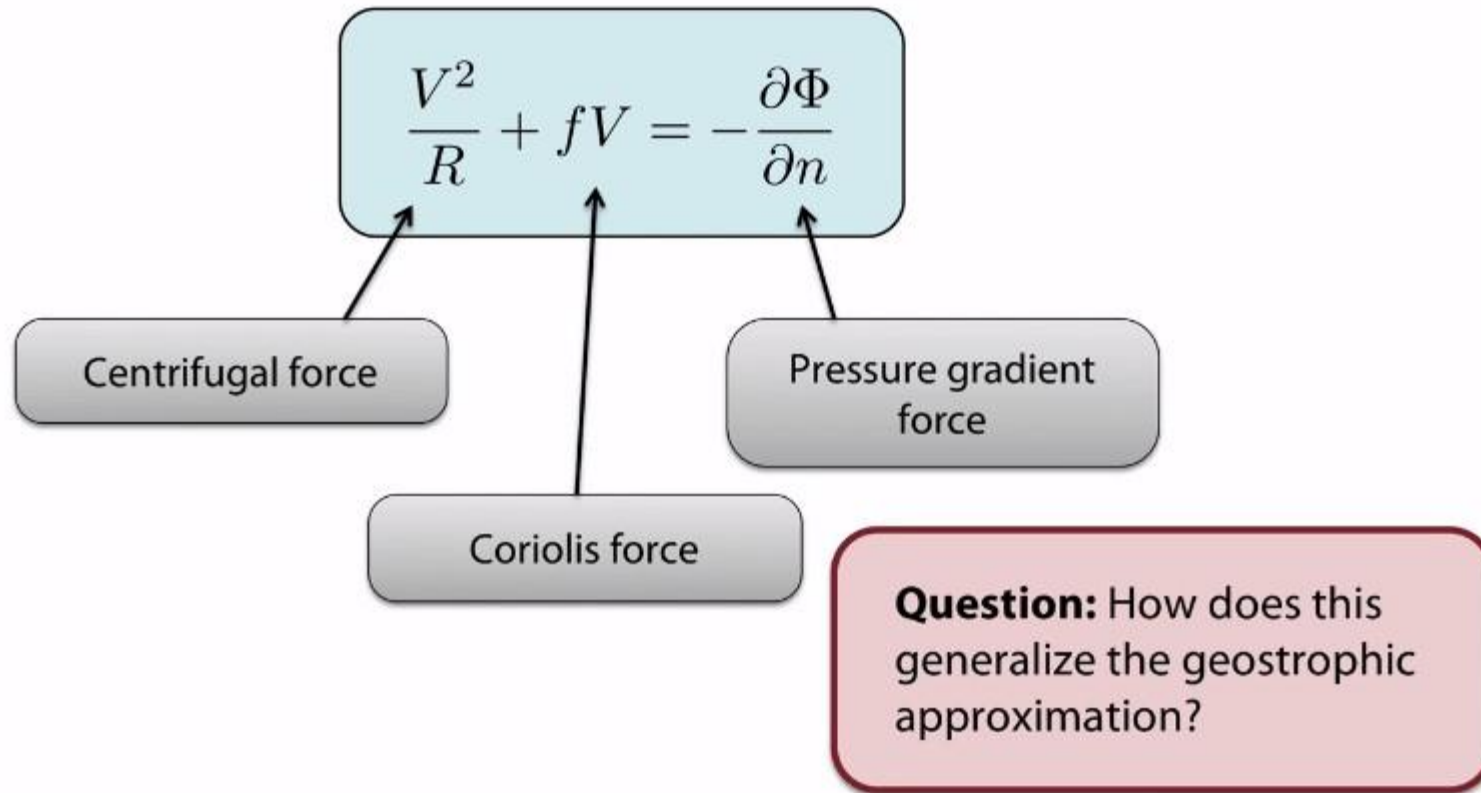
$$\frac{dV}{dt} = -\frac{\partial \Phi}{\partial s}$$
$$\frac{V^2}{R} + fV = -\frac{\partial \Phi}{\partial n}$$

Balance of forces parallel to flow.

Balance of forces normal to flow.

# Momentum Equation

One diagnostic equation for curved flow:



# *Geostrophic Flow*

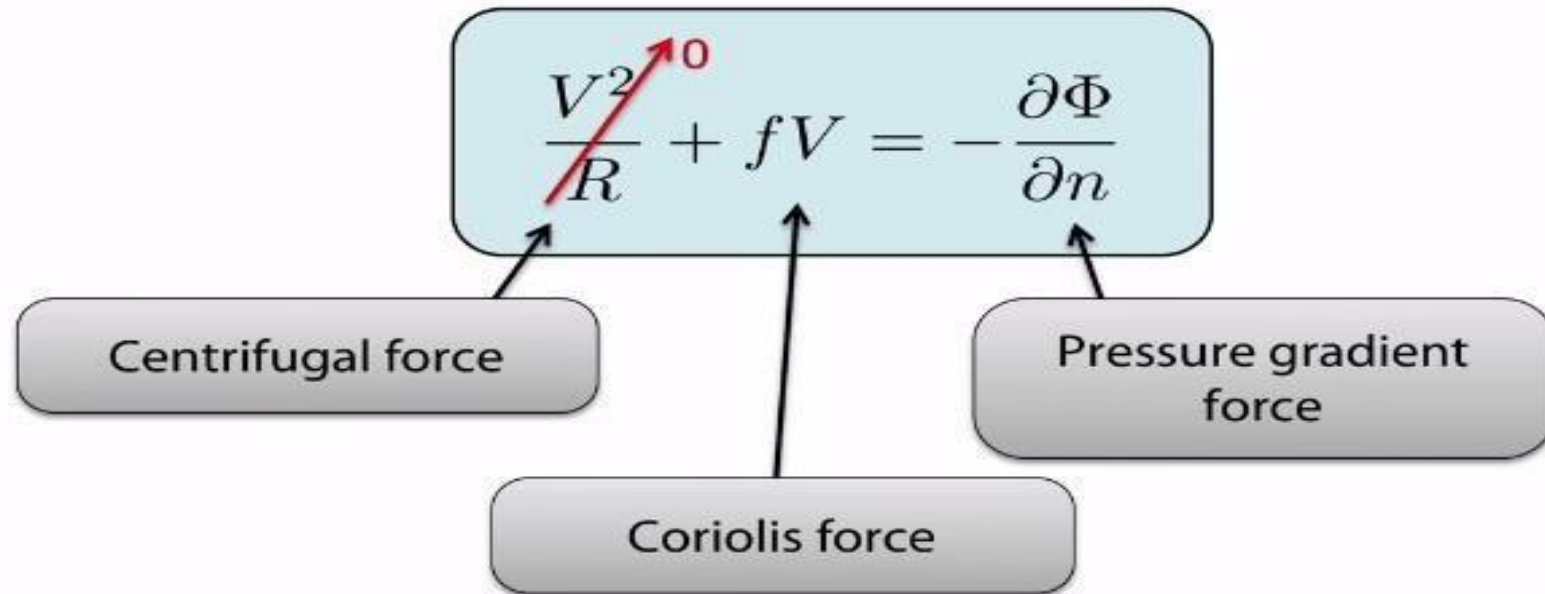
---

- Flow in a straight line ( $R \rightarrow \pm \infty$ ) parallel to height contours is referred to as *geostrophic motion*.
- In geostrophic motion the horizontal components of the Coriolis force and pressure gradient force are in exact balance.

# Geostrophic Balance

**Geostrophic balance:** Coriolis and pressure gradient force in exact balance (equal magnitude, opposite direction).

Flow is parallel to contours: Flow follows a straight line, so the radius of curvature ( $R$ ) becomes infinite.



# ***Geostrophic Balance***

***In Natural Coordinates***

Geostrophic balance in  
natural coordinates

$$fV_g = -\frac{\partial\Phi}{\partial n}$$

# ***Cyclostrophic Flow***

**Definition: Cyclostrophic flow** describes ***steady*** flows where centrifugal force is balanced by pressure gradient force, and Coriolis force is ***largely negligible***.



# ***Cyclostrophic Flow***

Normal component of momentum equation

$$\frac{V^2}{R} + fV = -\frac{\partial \Phi}{\partial n}$$

Cyclostrophic flow arises when  $V$  is large or  $R$  is small. In either case, the centrifugal force term is much larger than Coriolis force.

**Question:** When might these conditions occur?

## ***Cyclostrophic Flow***

Cyclostrophic flow arises when  $V$  is large or  $R$  is small. In either case, the centrifugal force term is much larger than Coriolis force.

**Tornadoes:** 100 meter radius, winds up to 50 m/s

**Dust Devils:** 1 – 10 meter radius, winds up to 25 m/s

Both of these phenomena feature small radii and (relatively) strong winds.

# *Inertial Flow*

- If the geopotential field is uniform on an isobaric surface so that the horizontal pressure gradient vanishes reduces to a balance between Coriolis force and centrifugal force

# Inertial Flow

**Definition: Inertial flow** describes **steady** flows where centrifugal force is balanced by Coriolis force and pressure gradients are **largely negligible**.

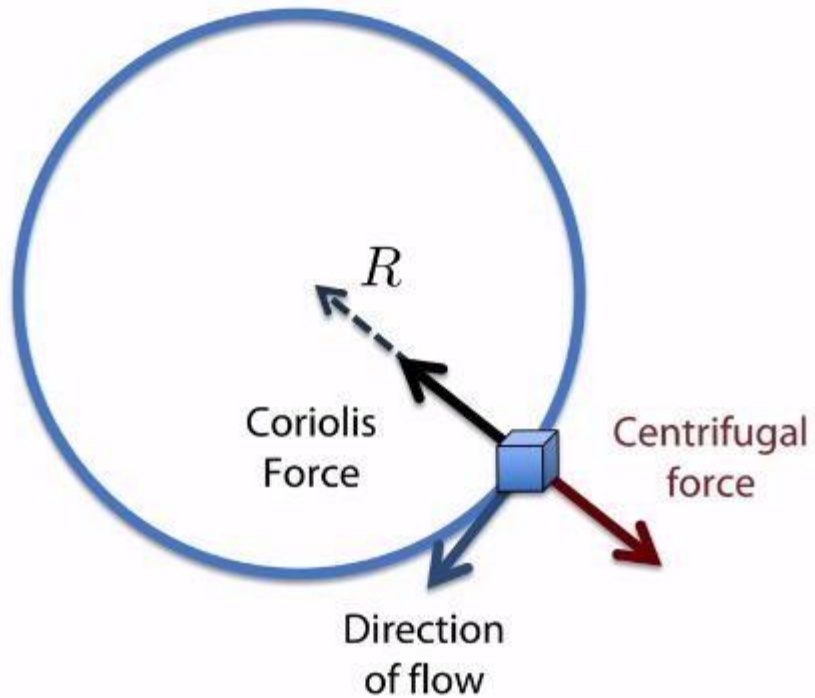
$$\frac{V^2}{R} + fV = -\frac{\partial \Phi}{\partial n} \overset{0}{\cancel{\partial n}} \Rightarrow \text{Solutions } V = 0 \overset{\text{Uninteresting}}{\cancel{V = 0}}$$

and  $V = -fR$

Since  $V > 0$ , must have  $R < 0$

# Inertial Flow

$$V = -fR$$



Inertial motion is always anti-cyclonic (clockwise in northern hemisphere).

Since the circumference of the circle is

$$C = 2\pi R$$

The period of rotation is

$$P = \left| \frac{C}{V} \right| = \frac{2\pi R}{fR} = \frac{\pi}{\Omega |\sin \phi|}$$

But since  $\Omega$  is rotation rate of the Earth

$$P = \frac{\frac{1}{2} \text{ day}}{|\sin \phi|}$$

# Gradient Flow

Momentum equation in natural coordinates, component form

$$\frac{DV}{Dt} = -\frac{\partial\Phi}{\partial s}$$

$$\frac{V^2}{R} + fV = -\frac{\partial\Phi}{\partial n}$$

So far we have investigated:

- Balance between pressure gradient force and Coriolis (**Geostrophic Balance**).
- Balance between pressure gradient force and Centrifugal force (**Cyclostrophic flow**).
- Balance between centrifugal force and Coriolis force (**Inertial flow**).
- We now consider balance between all three terms in the normal momentum equation (**Gradient flow**).

# *The Gradient Wind Approximation*

- Horizontal frictionless flow that is parallel to the height contours so that the tangential acceleration vanishes ( $DV/Dt = 0$ ) is called *gradient flow*.
- Gradient flow is a three-way balance among the Coriolis force, the centrifugal force, and the horizontal pressure gradient force.

# Gradient Flow

$$\frac{V^2}{R} + fV = -\frac{\partial\Phi}{\partial n}$$



$$V^2 + fRV + R\frac{\partial\Phi}{\partial n} = 0$$

Quadratic Equation



$$V = \frac{1}{2} \left[ -fR \pm \sqrt{(-fR)^2 - 4R\frac{\partial\Phi}{\partial n}} \right]$$



$$V = -\frac{fR}{2} \pm \sqrt{\frac{(fR)^2}{4} - R\frac{\partial\Phi}{\partial n}}$$

Observe this is simply a quadratic equation in  $V$

Gradient flow velocity



Considering the geostrophic balance the above equation can be written as

$$\mathbf{V} = -\frac{fR}{2} \pm \left( \frac{f^2 R^2}{4} + fRV_g \right)^{1/2}$$

# Reading assignment

- *Thermal wind*
- *a geostrophic wind*

....END

Thanks !

# Exercise

1. Which of the following is NOT a major atmospheric circulation cell?
  - A) Polar cell
  - B) Hadley cell
  - C) Ferrel cell
  - D) Walker cell
2. What is viscous force?
  - A) A force that is proportional to the temperature of the fluid
  - B) A force that is proportional to the pressure of the fluid
  - C) A force that is proportional to the velocity of the fluid
  - D) A force that is proportional to the density of the fluid
3. Which of the following factors can increase the magnitude of viscous forces in the atmosphere?
  - A) Increase in air temperature
  - B) Increase in air pressure
  - C) Increase in air density
  - D) Increase in wind speed

1. Which of the following factors affects the magnitude of the gravitational force in the atmosphere?

A) Altitude

C) Longitude

B) Latitude

D) Wind speed

2. Which of the following statements is true regarding the conservation of energy in atmospheric dynamics?

A. Energy can be created or destroyed

B. Energy can only be converted from one form to another

C. Energy can be created from nothing

D. Energy can be destroyed completely

3. What is the name of the phenomenon that occurs when the centrifugal force and the Coriolis force are in balance?

A) Equilibrium

C) Hydrostatic balance

B) Geostrophic balance

D) Inertial flow

1. Hydrostatic equilibrium occurs when:

A) Large air masses are moving either up or down

B) The force of gravity and the vertical pressure gradient both act to push air upward.

C) The force of gravity and the vertical pressure gradient have equal values and oppose each other.

D) The force of gravity and the vertical pressure gradient both act to push air downward.

2. What is the name of the phenomenon that occurs when the centrifugal force and the Coriolis force are in balance?

A) Equilibrium

C) Hydrostatic balance

B) Geostrophic balance

D) Inertial flow

1. .... Is a flow that occurs when Coriolis force and pressure gradient force are in exact balance
  - A. Gradient flow
  - B. Inertial flow
  - C. Cyclostrophic flow
  - D. Geostrophic flow
2. A coordinate system used to define balanced flow conditions is
  - A. Spherical Coordinate System
  - B. Cylindrical Coordinate System
  - C. Natural Coordinate System
  - D. Projected Coordinate System
3. .... Is a flow that occurs centrifugal force and pressure gradient force are in exact balance
  - A. Gradient flow
  - B. Cyclostrophic flow
  - C. Inertial flow
  - D. Geostrophic flow